

ECE:5995 Applied Machine Learning

Unsupervised Scene Categorization with RGB-D data

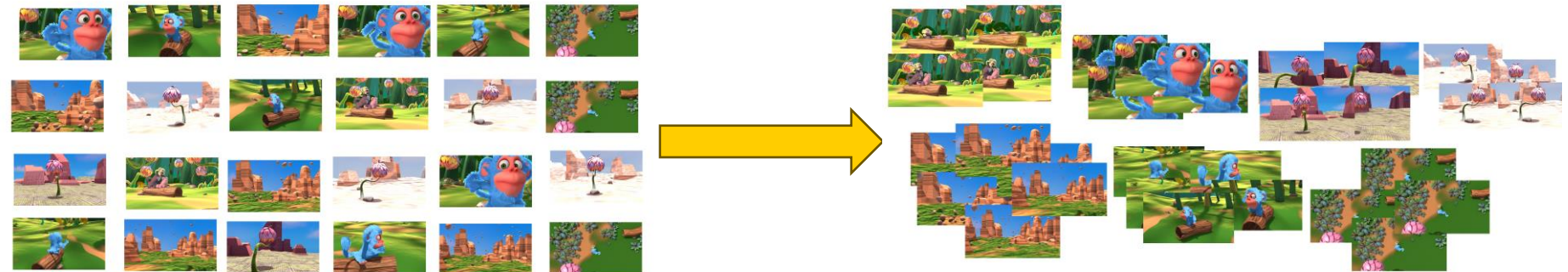
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Overview

- We wanted to be able to cluster scenes captured from a short film using RGB-D data with no prior knowledge of labels.

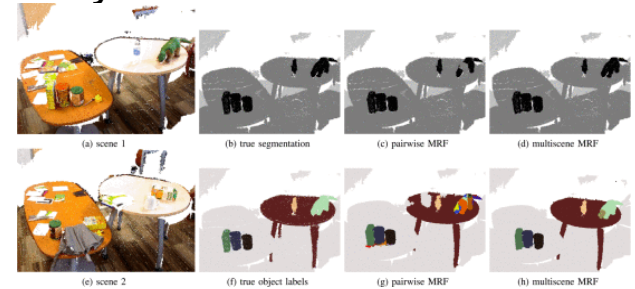


- By combining both RGB and depth information, we intend to enhance scene representation and understanding, particularly focusing on spatial relationships and object geometries.
- Clustering scenes by utilizing depth could be useful in systems that capture/utilize 3D data such as Augmented/Virtual Reality or Robotics & Navigation.



Background

Previous research in clustering RGB-D data has mainly focused on tasks such as 3D reconstruction and plane segmentation.



[Yurong You, Katie Luo, Cheng Perng Phoo "Learning to Detect Mobile Objects from LiDAR Scans Without Labels", 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR\), pp.1120-1130, 2022.](#)

Additionally, related work has explored the use of RGB-D cameras for unsupervised object discovery and scene understanding.

- The goal is to enable robots to learn about objects in an unsupervised manner by detecting changes in the environment over time.
- Our project differs in focus from this work by concentrating specifically on scene categorization rather than object discovery and motion tracking.
- Insights from these related works inform our understanding of the capabilities and challenges associated with processing RGB-D data for scene analysis.



Approach

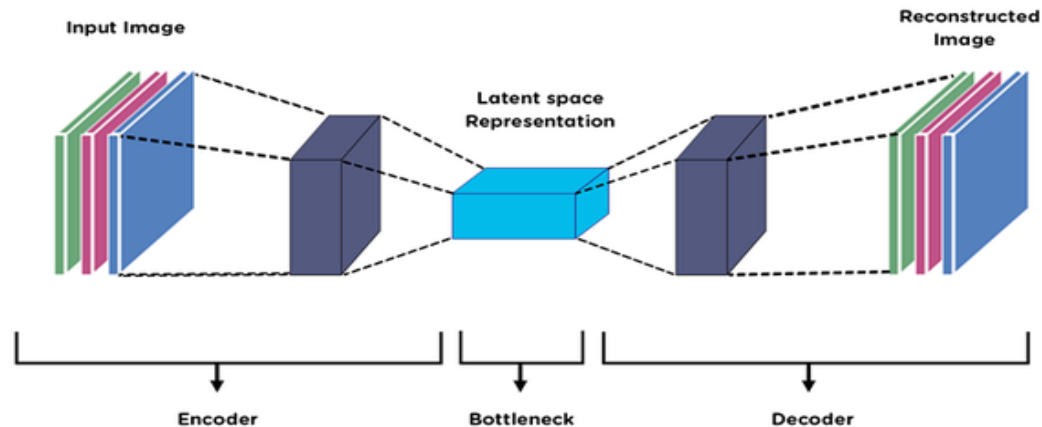
- Unsupervised Clustering
 - Use an Auto-Encoder so we can later use it to extract key features from scenes
 - Once the key features have been extracted, perform K-Means clustering algorithm
- Monkaa Dataset
 - ~35,000 stereo frames with ground truth for disparity maps
 - Memory limitations allowed us to use about half of the scenes
- Training & Evaluation
 - Shuffle weights when training, but extract features from data in-order so we can compare with ground truth data array we created (since # of scenes were known)
 - Saved weights depending on the lowest MSE loss each epoch
 - Used Adam optimizer for its adaptive learning rate



Approach

Constructing our Neural Network

- The encoder consists of several convolutional layers designed to compress the input data into a smaller latent space representation.
 - This latent space is what we use for the clustering
- Varied the size of the number of layers, encoding output, as well as different loss functions for hyperparameter tuning while training.

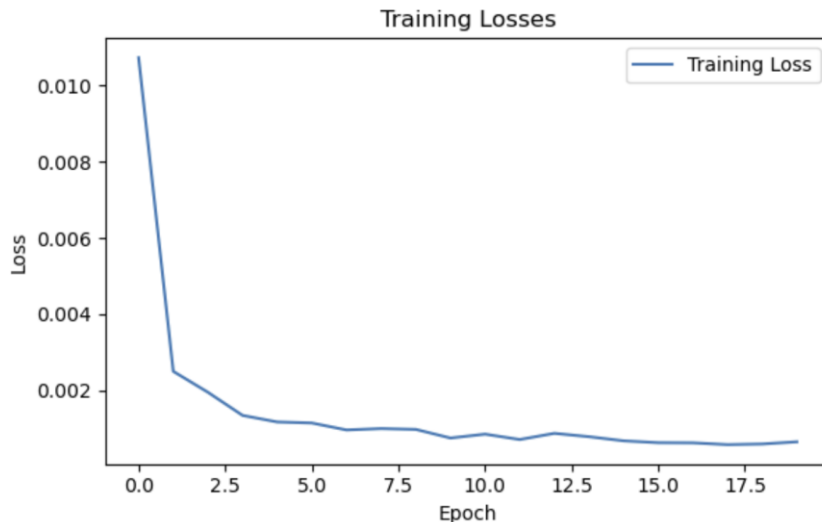


Results

Achieved 100% accuracy for clustering the data into their respective scenes without knowledge of any labels

Used ARI & AMI metrics to compare our cluster array with ground truth data

- A score of 1 indicates perfect similarity
- A score of 0 or negative indicates random or dissimilar clusterings



```
[117]: # Assuming ground_truth_labels is the array with ground truth labels
# Assuming clusters is the array with the clustering results

# Calculate Adjusted Rand Index (ARI)
ari_score = adjusted_rand_score(ground_truth, clusters)

# Calculate Adjusted Mutual Information (AMI)
ami_score = adjusted_mutual_info_score(ground_truth, clusters)

print("Adjusted Rand Index (ARI):", ari_score)
print("Adjusted Mutual Information (AMI):", ami_score)
```

Adjusted Rand Index (ARI): 1.0
Adjusted Mutual Information (AMI): 1.0



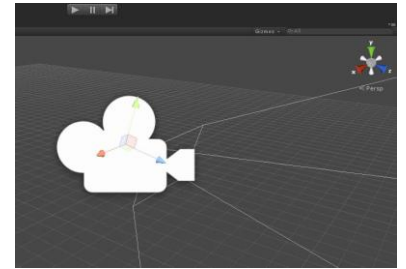
Adjusted Rand Index (ARI): 0.961499703374664
Adjusted Mutual Information (AMI): 0.9673005080338367

- Achieved ~0.96 score
- Some scenes had mismatches

[illegible]

Conclusion

- We believe including the depth information was helpful in the clustering of the data
- Future work
 - Include camera data which was provided by the dataset (transformation, rotation, etc)
 - Look into using 3D convolutions
 - Pros
 - Could be very helpful as they are designed to capture the spatial and temporal dimensions of the data
 - Since the scenes in our data have slight variations from frame-to-frame, a 3D convolution may be able to learn these temporal differences
 - Cons:
 - computationally expensive & requires substantial amounts of data



Thank you

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Members

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Github Repo

<https://github.com/Nick-Hageman/RGBD-Unsupervised-Clustering>

Citations

[Yurong You, Katie Luo, Cheng Perng Phoo "Learning to Detect Mobile Objects from LiDAR Scans Without Labels", 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR\), pp.1120-1130, 2022.](#)

